

Module 02: Agents in High School

Learning Objectives

1. Define **Agents** within the context of High School.
2. Analyze how Agents interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Agents.

Introduction

This module explores **Agents**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Agents is a critical component of the 8-part Active Inference spine, bridging the gap between Systems and Perception.

Key Concepts

1. Agents as a Markov Blanket Boundary

How does Agents define the boundary between the agent and the environment?

2. Generative Models of Agents

What parameters involved in Agents must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Agents drive the perception-action loop?

Applications

In High School, we see Agents manifest in: * **Specific Example 1:** A self-driving car is a technology agent running Active Inference: its LIDAR and cameras provide sensory

observations, its onboard neural networks maintain a generative model of the road environment (predicting where pedestrians will walk, how traffic will flow), and its steering and braking actuators execute action policies that minimize expected free energy -- keeping the car safely within predicted lane boundaries. * **Specific Example 2:** A recommendation algorithm on YouTube or TikTok acts as an AI agent performing Active Inference on your preferences: it maintains a generative model of what videos you will watch next, updates that model every time you watch, skip, or like a video (prediction error), and selects the next recommendation to minimize the surprise of you scrolling past -- which is why your feed feels increasingly tailored the more you use it.

Conclusion

Understanding Agents allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.